Status	Finished
Started	Tuesday, 20 October 2026, 3:51 PM
Completed	Tuesday, 20 October 2026, 4:00 PM
Duration	8 mins 35 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

a. Heap sort

b. Selection sort

c. Quick sort

d. Merge sort

The correct answer is:
Merge sort

Question **2**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

a. It increases the number of comparisons needed

b. It has no effect on the search operation

c. It allows for faster searching

d. It makes the search operation slower

The correct answer is: It allows for faster searching

Question **3**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

- a. Backtracking
- b. Divide and conquer
- c. Greedy method
- d. Dynamic programming

The correct answer is:
Divide and conquer

Question **4**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

- a. Bubble Sort is difficult to implement
- b. Bubble Sort does not guarantee sorted order
- c. Bubble Sort is less efficient for large lists
- d. Bubble Sort cannot handle duplicate elements

The correct answer is: Bubble Sort is less efficient for large lists

Question **5**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

- a. It repeatedly finds the minimum element and moves it to the sorted part of the list
- b. It compares adjacent elements and swaps them if necessary
- c. It builds a sorted array one element at a time
- d. It divides the list into two halves, sorts each half, and then merges them

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **6**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

a. Bubbling up the smallest element
b. Sorting the entire list in one pass
c. Dividing the list into halves
d. Bubbling up the largest element to its correct position

The correct answer is: Bubbling up the largest element to its correct position

Question **7**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

a. They are combined to form a single sorted list
b. They are split again into smaller sublists
c. They are compared element by element
d. They are discarded

The correct answer is: They are combined to form a single sorted list

Question **8**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

a. Sorting data in a non-sequential manner
b. Using the divide-and-conquer approach
c. Being the most efficient sorting algorithm
d. Bubbling up the largest element to its correct position with each pass

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **9**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

a. It slows down the process
b. It makes it easier to select items based on their relationship to the rest of the items
c. It complicates the selection of items
d. It makes the data unsorted

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **10**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

a. sort()
b. sorted()
c. arrange()
d. order()

The correct answer is: sort()

Question **11**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Comparing each element with the others
b. Combining sorted sublists
c. Sorting the entire list sequentially
d. Dividing the input into smaller subproblems

The correct answer is: Dividing the input into smaller subproblems

Question **12**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. order()
- b. sorted()
- c. sort()
- d. arrange()

The correct answer is: sorted()

Question **13**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

- a. Sorting helps in finding duplicates quickly
- b. Sorting decreases the efficiency of selection operations
- c. Sorting is rarely used in programming
- d. Sorting makes it harder to search for items

The correct answer is: Sorting helps in finding duplicates quickly

Question **14**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

- a. It recursively sorts subproblems
- b. It uses the heap data structure
- c. It bubbles up the largest element in each pass
- d. It divides the list into sublists

The correct answer is: It bubbles up the largest element in each pass

Question **15**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

a. It sorts data out of the box efficiently
b. It requires external libraries
c. It is less efficient than custom sorting algorithms
d. It only works with integer arrays

The correct answer is: It sorts data out of the box efficiently